

Rank - Nullity Theorem

• Let $A: \mathbb{R}^n \rightarrow \mathbb{R}^m$, rank + nullity = n

$$\dim \text{Null}(A) + \dim \text{Col}(A) = n$$

Proof

① Let's consider nullity $(A) = k$

② We have a basis $\{u_1, \dots, u_k\}$ of $\text{null}(A)$

③ Extend $\{u_1, \dots, u_k\}$ to a basis of $\mathbb{R}^n$

$$\{u_1, u_2, \dots, u_k, w_1, \dots, w_{n-k}\}$$

④ Now let's take $\{Aw_1, \dots, Aw_{n-k}\}$

⑤ Now we claim that this set is a basis of column space of A . For that we show that it is a spanning set and an independent.

⑥ Let's apply A to the earlier extended set.

$$x = \sum \alpha_i u_i + \sum \beta_j w_j$$

$$Ax = \alpha_i \sum A u_i + \beta_j \sum A w_j$$

null space

$Ax \rightarrow$ spans the whole col space
Hence $\sum A \beta_j w_j$ spans the whole column space so $\{w_1, \dots, w_{n-k}\}$ is a span for $\text{col}(A)$.

⑦ Suppose $\sum \beta_j A w_j = 0$ then by definition $\sum \beta_j w_j$ must be in the null space of A .

$$\sum \beta_j w_j = \sum \alpha_i u_i \quad \text{for some } \alpha_i$$

$$\sum \beta_j w_j - \sum \alpha_i u_i = 0$$

Now since $\{u_i^*, w_j^*\}$ form a basis for $\mathbb{R}^n$, all α_i, β_j are 0 and hence, the w_j set is independent.

⑧ Hence the $\text{rank}(A) = n - k$

Rank of Transpose

$$\text{rank}(A) = \text{rank}(A^T)$$

Read about proof.

Number of solutions

$$A x = B$$

- Rank n $\rightarrow$ surjectivity (solution exists)
injectivity (unique solⁿ)
- Rank $< n$ $\rightarrow$ surjective (solution exists)
 & not injectivity (infinitely many)
 $B \in \text{Im}(A)$

- Rank $< n \rightarrow$ no solution $\neg$ surjectivity
 $\&$
 $B \notin \text{Im}(A)$

Exercises

①
$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix}$$

LU decomp. $\rightarrow$ $R_2 \rightarrow R_2 - 2R_1$
 $R_3 \rightarrow R_3 - 4R_1$

$\Rightarrow$
$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{bmatrix} \rightarrow R_3 \rightarrow R_3 - 3R_2$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix}$$

(L)

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

(U)

Determinants / Invertibility

⇒ A square matrix is invertible when there is A^{-1} such that

$$AA^{-1} = A^{-1}A = I$$

⇒ $GL_n(\mathbb{R})$ is a group of invertible $n \times n$ matrices over $\mathbb{R}$.

⇒ There is a similarity relationship

$$A' \sim A$$

$$A' = P^{-1}AP$$

• Prove that it is a similarity

✓
① Reflexivity $A \sim A$

$$A = I^{-1} A I$$

✓
② Symmetry $A \sim A' \Rightarrow A' \sim A$

$$A' = P^{-1} A P$$

$$P A' = A P$$

$$\boxed{P A' P^{-1} = A}$$

✓
③ $A \sim B$ Transitivity
 $B \sim C$
 $\Rightarrow A \sim C$

$$\begin{array}{lcl} A = P^{-1} B P & \xrightarrow{\quad} & A = P^{-1} Q^{-1} C Q P \\ B = Q^{-1} C Q & \hookrightarrow & A = (Q P)^{-1} C (Q P) \end{array}$$

Determinant

$$\det A : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$$

$\Rightarrow$ Multilinear in rows/cols.

Calculation of determinant

a Minor $A_{i,j}$ as $(n-1) \times (n-1)$ matrix
where we have removed i, j from the
matrix.
row col

$$\det(A) = \sum_{j=1}^n a_{ij} (-1)^{i+j} \det(A_{ij})$$

or we can iterate over
a row.

$$\Rightarrow \det(A) = \det(A^T)$$

$\Rightarrow$ In a triangular matrix is the product of diagonal terms.

$$\Rightarrow \det(AB) = \det(A) \det(B)$$

Read about proof

$\rightarrow$ Use the fact that determinant of AB is a "multilinear" function.

$\rightarrow$ Then prove $\det(AB) = c \det(A)$

$\rightarrow$ Then assume $A = I$ & c comes out as $\det(B)$.

$$\Rightarrow \det(\tilde{A}) = \frac{1}{\det(A)}$$

Proof

Since $AA^{-1} = I$

$$\det(AA^{-1}) = 1$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

$\Rightarrow$ Similar matrices as same determinant

Proof

$$A' = P^{-1} A P$$

$$\det(A') = \det(P^{-1} A P)$$

$$\det(A^{-1}) = \det(P^{-1}) \cdot \underbrace{\det(A) \cdot \det(A)}_1$$

$$\det(A^{-1}) = \det(A)$$

⇒ Adding a linear combination of rows to a row, the determinant doesn't change.

Proof

$$\det(A) = D(r_1, \dots, r_n)$$

$$D(r_1, r_2, \dots, r_i + \sum_{j \neq i} \alpha_j r_j, \dots)$$

$$\Rightarrow D(r_1, r_2, \dots) + \sum \alpha_j D(\dots \cancel{r_j} \dots)$$

would be seen as other row so 0

The following are equivalent

① A is invertible.

② $\det(A) \neq 0$

③ cols & rows of A are linearly ind.

④ $\text{Null}(A) = \{0\}$

⑤ $AX = B$ has a unique solution for any B .